

Company : Synapsis.id
Job Title : Firmware Engineer
Period : June 2025 – July 2026

- **Overview**

In June 2025 until July 2026, I was working in a tech startup called Synapsis.id located in East Jakarta, Indonesia. I was involved in both development team and research team for a fleet management system (FMS) product, responsible in maintaining the existing firmware and also conducting intensive research for the next version of the firmware.

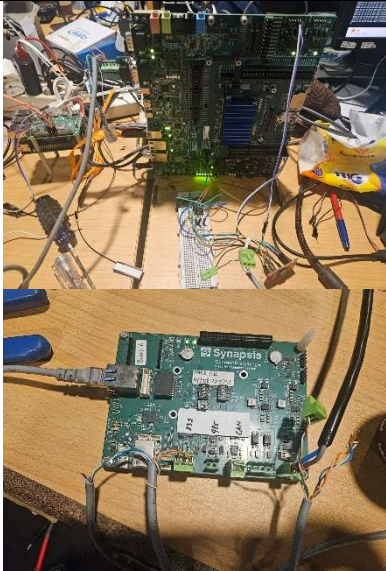
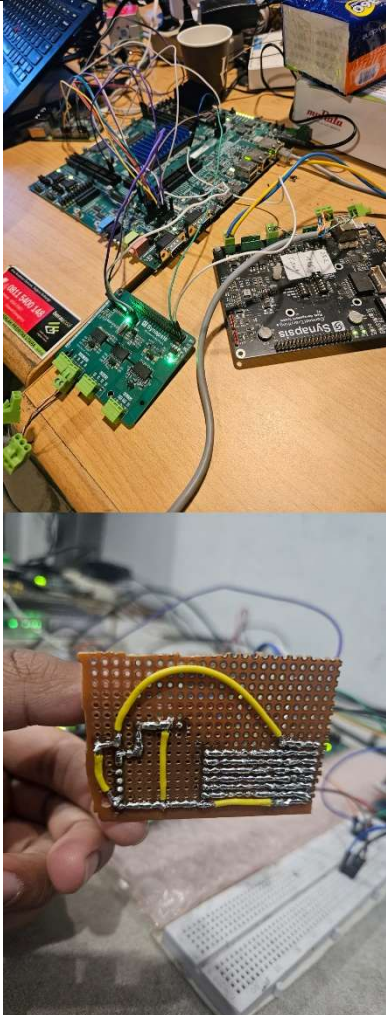
- **Core Responsibilities**

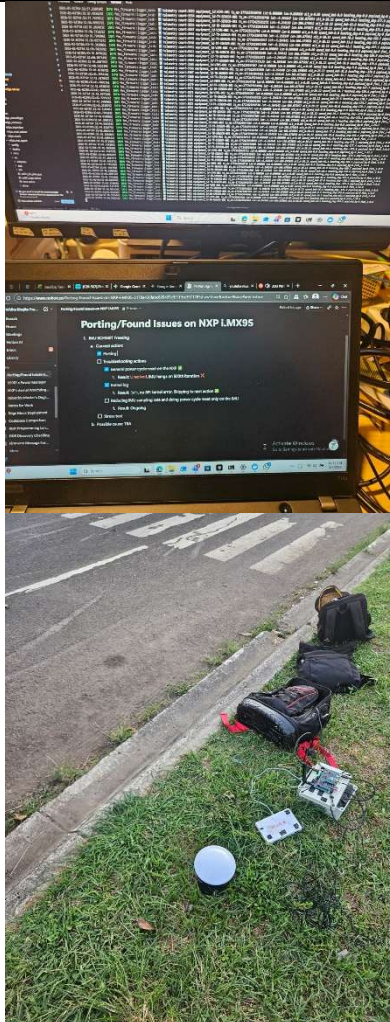
- Maintain the IoT-based fleet management system (FMS) firmware for mining fleets.
- Explore new devices, features, and tools to support the FMS development.
- Create technical documentations along the development processes.
- Explore new methods and algorithms for mitigating GNSS outage issue as a lead firmware researcher in the research team.
- Involve in the static and dynamic testing phases in order to verify the FMS reliability before being deployed to the user.
- Support the field engineer during the firmware deployment phase.
- Collaborate in a cross-functional team discussion in order to create a robust system.
- Troubleshoot issues related to the developed FMS.

- **Achievements**

- Involved in 2 firmware deployment periods.
- Acted as a lead firmware researcher in the research team to develop GNSS outage mitigation system by fusing sensing data from GNSS, IMU, and ECU.
- Created a rust-based GNSS outage mitigation system that lasted 10 hours of stress test and around 1 month of field test.
- Proposed a method for testing the I2C device interfacing firmware without using the real hardware by creating an application that can continuously stream dummy data that mimic the real data frame from the actual I2C device in an adjustable frequency.

- **Documentation**

No.	Activity	Documentation
1.	Firmware functionality test using NXP iMX8M as the main processor and Raspberry Pi CM4 as the dummy data streamer.	
2.	Debugged the new circuit board manufactured by the electronic team and also created a prototype board to test the firmware in advance.	

3.	Intensively involved in the static testing phase and the dynamic testing phase. Detected and solved issues found during the static testing phase, such as un-monotonic sensor data timestamp, hanging IMU data stream, and linux OS booting issues. During the dynamic testing phase, involved in sensor data sampling activities by driving the FMS around Halim Perdanakusuma Airport, East Jakarta while running the firmware, and also assisted in GNSS base station set up.	
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4.	Involved in the firmware deployment process.	